# Forecasting Takeover in LLM Agent Populations: Preregistered Tests of Policy Emulators and the Limits of Early Warning*

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**Abstract**

A small committed minority can tip a population of LLM agents into a new convention, an abstraction of a few compromised agents hijacking a swarm's norms. Large-population estimates of the required *critical mass* rely on emulators driven by policies extracted from the model, yet these forecasts have not been tested prospectively. In an LLM naming game we run two preregistered live forecasting tests (DeepSeek-v4-flash; 67 populations; 121k decisions) and diagnose forecast errors against exact log-probability policies of three open-weight models. (i) Live decisions depart only slightly from the offline policy samples (calibrated test $p \approx 0.03$; estimated RMS difference $\approx 0.01$), and substituting live-estimated probabilities changes the emulated critical mass by $\leq 0.011$. (ii) A preregistered emulator built from a policy pooled across word pairs failed (2/5 levels outside 95% predictive intervals) and had no skill over the base rate (Brier 0.22 vs 0.22). Exact policies show why: averaging the transition policy across contexts yields a transition far sharper than the corresponding mixture of populations (flip-probability error up to 0.37), and the smoothed estimators we tested bias critical mass low by up to 60% at small sample sizes. (iii) A preregistered target-conditional emulator correctly predicted the direction of a large target asymmetry (10/10 vs 2/10 takeovers, $p = 3.6\times10^{-4}$) and improved the Brier score to 0.09; its quantitative calibration is only weakly tested. (iv) In emulation, surrogate-calibrated early-warning indicators showed no consistent incremental discrimination once population level and trend were matched. We report all failed predictions and withdrawn claims.

## 1. Introduction

Populations of interacting LLM agents can self-organise into collective behaviour that no single-agent evaluation anticipates; the July 2026 Hugging Face incident is a recent example [16]. A concerning failure mode is *norm hijacking*: a handful of agents that never update can, through ordinary repeated interaction, pull a population onto a convention of their choosing. In the classical committed-minority model this happens once the adversarial fraction exceeds a critical mass [8, 9], and recent work reproduces such tipping points with LLM agents [1–5].

An overseer wants to know *how many agents an adversary must control* and *whether a takeover can be seen coming*. Because live experiments are expensive, large-population answers rely on **emulators**: simulations driven by policies extracted from the LLM [2–4]. Prior work validated per-pair, log-probability emulators against direct LLM simulations [2, 3] and mapped directional, convention-dependent critical masses across models [3]. What has not been done is to **forecast prospectively**: freeze an emulator's numerical predictions, then test them on fresh live populations, and diagnose the errors. This matters most for black-box APIs that expose no log-probabilities, where the policy must be estimated from samples and choices about aggregation and smoothing are unavoidable.

**Threat model.** An adversary controls c of N agents (committed fraction $p = c/N$) that always play a target convention B, or poisons agents' memories at a hidden rate q(t). The adversary may choose which convention to push. The defender can query the model offline and observe the population's aggregate behaviour.

**Contributions.**

1. **Two preregistered forecasting tests.** An emulator built from a policy pooled across word pairs *failed* and had no skill over the base rate; a target-conditional emulator correctly forecast a large directional asymmetry and improved proper scores.
2. **A decision-level check of the policy-following assumption for a sampled (no-logprob) API model**, with calibrated inference, power analysis, and the consequence for the forecast quantity.

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3. **A controlled diagnosis of forecast error against exact policies**: aggregation across contexts versus sampling and smoothing, with bias, RMSE and interval coverage of the critical-mass estimate.
4. **A level-matched, surrogate-calibrated evaluation of early-warning indicators** for an emulated memory-poisoning driver, as a bounded secondary result.

# 2. Related Work

**Conventions and committed minorities in LLM populations.** Ashery et al. [1] showed that LLM populations form conventions that committed minorities can overturn, noting that the tipping point depends on the conventions involved. Flint et al. extracted per-pair log-probability policies, validated the resulting model against direct LLM simulations, and studied group size [2]. Their *indirect tipping* study [3] showed quantitatively that critical masses are directional and convention-dependent across four models, forming a directed graph over equilibria. Passaro et al. [4] reused those per-pair policies to reach $10^3$–$10^5$ agents across topologies and excluded conditions whose convention is unstable with zero committed agents. De Marzo et al. [5] report conformity-driven misalignment and hysteresis; De Nobili et al. [6, 7] give statistical-physics descriptions of LLM populations. **Target dependence, emulator validation with log-probability policies, and the stability precondition are therefore established.** Our contribution is complementary: *preregistered* numerical forecasts tested on fresh live populations, a validation for a *sampled* policy from an API without log-probabilities, and a controlled account of how estimation and aggregation choices propagate into critical-mass forecasts.

**Early warning and multi-agent risk.** Critical slowing down [10, 11] is used to anticipate transitions, but conditioning on transitions inflates apparent performance [12]; indicators can also be learned [13]. Hammond et al. [14] catalogue multi-agent risks, and Makins et al. [15] show that per-instance monitors miss distributed attacks.

# 3. Methods

## 3.1 Environment and definitions

We adapt the naming game of Ashery et al. [1] (MIT licence). N = 24 agents start on convention A. Each round agents are paired by a uniformly random perfect matching, and each pair plays a coordination game (+100 if both choose the same word, −50 otherwise). An agent's prompt contains only its last five records (own word, partner's word, payoff); we remove the cumulative score of the original prompt so that no hidden state remains. Option order is shuffled per call, reasoning is disabled (0 reasoning tokens verified), and output is JSON with one resample on parse failure (0 unrecoverable failures in 121,423 live calls). c committed agents always play B.

The observable $f_t$ is the fraction of non-committed agents playing B. A population **flips** if the two-round mean of $f_t$ reaches 0.5 within 150 rounds: a *finite-horizon first-passage event* for a given N, horizon, initial state and event definition, not a statement about an asymptotic bifurcation. $\varphi(p)$ is the flip probability and the **critical mass** $p_c$ solves $\varphi(p_c) = ½$ (linear interpolation between integer c). Emulator flip probabilities are Monte Carlo estimates ($\geq 300$ runs; SE $\leq 0.03$; 1,000 runs, SE $\leq 0.016$, for frozen predictions). The flip event agreed with sustained adoption ($f_t \geq 0.9$ for 5 rounds) in 97–100% of live runs (§4.4).

The prompt is a deterministic function of the memory state s ($2^{10} = 1024$ states), word pair w and option order o. Because order is randomised per call, the population dynamics are driven by the order-marginalised policy $\pi(s, w) = E_o\, P(B \mid s, w, o)$; an emulator with the true $\pi(s, w)$ reproduces the live protocol up to Monte Carlo error.

## 3.2 Policy extraction and emulators

**API model (DeepSeek-v4-flash).** No log-probabilities are exposed, so $\pi$ is estimated by sampling: 8 samples per state cycling through six word pairs, plus 40–48 extra samples on near-threshold states, and pair-specific top-ups for the two pairs of Test 2. **Local models (Qwen3-8B, Gemma-3-12B, Mistral-7B, Llama-3.1-8B; 4-bit MLX community builds on an Apple M5 Pro).** We compute next-token probabilities of the two words after a forced `{"value": "` prefix, renormalised over the two answers and tempered to T = 0.7 ($\geq 97\%$ of mass on the two answers), for all 1024 states × 6 pairs × 2 orders. We call these **exact policies**: exact for this forced-prefix two-answer policy of these quantised builds, not for unrestricted generation. We compare *pooled* ($\pi$ averaged over pairs) vs *target-conditional* ($\pi$ for one pair) emulators, and *unsmoothed* maximum-likelihood estimates k/n vs *Jeffreys* $(k + ½)/(n + 1)$ and *Laplace* $(k + 1)/(n + 2)$ estimates.

### 3.3 Preregistered live tests

Before each live experiment we recorded SHA-256 hashes of the preregistration, code, policies and predictions (the documents are available from the author; hashes establish content identity, not independent chronology). **Test 1** used the pooled DeepSeek policy and counterbalanced word pairs by seed (pair allocation was therefore not balanced within every level of c). H1: at every level of c the live flip count lies inside the emulator's 95% predictive interval (PI), evaluated by an interval rule and a Holm-corrected binomial rule. **Test 2** (addendum) used target-conditional policies for "push Q" (A = M, B = Q) and "push M" (A = Q, B = M) with fixed words and fresh seeds. The primary H5 is directional: flips(push M) > flips(push Q) at c in {3, 4} (one-sided Fisher exact test); H1′ repeats the PI check. PIs were built from Jeffreys-Beta posterior draws of the policy, which §4.3 shows to be biased; we treat this as a limitation and add proper scores against baselines.

### 3.4 Monitoring evaluation (emulation)

The hidden driver is memory poisoning: each round each agent's new record is forged (partner recorded as playing B) with probability q(t). In emulation with N = 48 (not the N = 24 of the live tests), TOWARD runs ramp q from 0 to $1.6 \cdot q_c$ over 300 rounds, and 4,700 non-tipping null runs cover stationary, stop-short and away-ramp schedules. Segments ending L in {10, 25, 50, 100} rounds before the tip are matched 1:1 to null segments on final level and trend. Indicators are variance and lag-1 autocorrelation (AC1) converted to surrogate-calibrated z-scores, plus level, trend, recent slopes and an oracle that observes q. Incremental value is the grouped cross-validated logistic AUROC of level + trend with and without the indicators. Uncertainty uses a **cluster bootstrap over connected components of matched pairs that share a null run**, and a refit bootstrap for the preregistered comparisons (H4: indicators add ≤ 0.02 AUROC). This is a retrospective discrimination task on selected pre-tip segments, not a prospective alarm.

## 4. Results

### 4.1 Live decisions versus the extracted policy

We reconstructed the context of 121,375 first-attempt live decisions and compared them with 38,948 offline samples of the same model, cell by cell. Using cells with ≥ 5 decisions on both sides (579 cells with outcome variation, covering 64,002 live decisions), a homogeneity test **calibrated by parametric bootstrap** rejects equality (G = 680, p = 0.026; p = 0.016–0.13 across strata: test 1 or 2 only, early or late rounds). The difference is small: the estimated root-mean-square difference between true live and extracted probabilities is 0.006 (95% CI 0–0.029; order-conditional cells 0–0.040), and live agents chose B slightly more often (signed +0.005). The test is well powered (89% power for a 0.03 shift). For the forecast quantity, replacing extracted with live-estimated probabilities in the 446 states with ≥ 5 live decisions changes the emulated critical mass from 0.146 to 0.145 (pooled), 0.184 to 0.182 (push Q) and 0.117 to 0.106 (push M). **Live agents deviate detectably but slightly from the extracted policy, and the deviation has little effect on critical-mass forecasts.** (A uniform 0.05 shift in all choice probabilities would move it to 0.073 or 0.178.)

### 4.2 Preregistered test 1: a pooled emulator fails near threshold

The pooled-policy emulator predicted $p_c = 0.135$. Table 1 compares its predictions with 32 live populations.

Table 1: **Table 1.** Preregistered test 1 (DeepSeek-v4-flash, N = 24). k = number of populations that flipped. H1 requires k to lie inside the emulator's 95% predictive interval.

| c | p | Live flips | Emulator φ | 95% PI (k) | H1 |
|---|---|---|---|---|---|
| 2 | 0.083 | 1/4 | 0.00 | [0, 0] | **fail** |
| 3 | 0.125 | 4/8 | 0.11 | [0, 5] | pass |
| 4 | 0.167 | 5/8 | 0.97 | [7, 8] | **fail** |
| 5 | 0.208 | 8/8 | 1.00 | [8, 8] | pass |
| 8 | 0.333 | 4/4 | 1.00 | [4, 4] | pass |

**H1 is rejected** under both preregistered rules. The c = 4 failure is robust ($P(k \leq 5 \mid \varphi = 0.962, n = 8) \approx 0.003$); the c = 2 failure rests on a single run. The pooled forecast had **no skill over a constant base-rate forecast** (Brier 0.219 [0.090,

0.358] vs 0.215; log-loss 0.82 vs 0.62). *Post hoc*, outcomes at c = 2–4 tracked the committed word (push M 5/5, push Q 0/5; Fig. 1), and re-predicting each run with its own pair's unsmoothed policy reduced the Brier score to 0.09. Because pair allocation also varied across levels, the failure is not an isolated test of pooling; §4.3 isolates pooling with exact policies.

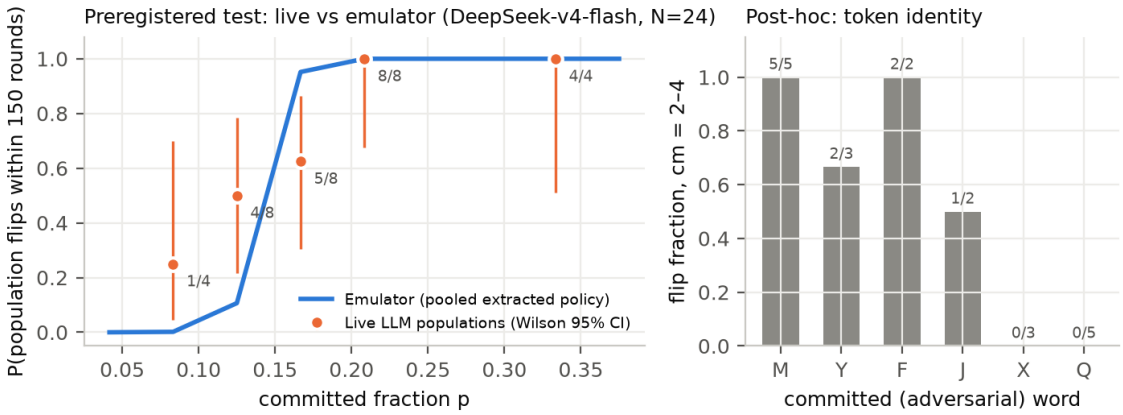



Figure 1: **Figure 1.** Preregistered test 1. *Left:* live flip fraction (Wilson 95% CI) vs the pooled-policy emulator. *Right (post hoc):* flips at c = 2–4 split by the committed word.

## 4.3 Diagnosing forecast error with exact policies

**Precondition.** Llama-3.1-8B does not hold a convention without adversaries (P(B | five successful A rounds) up to 0.95; spontaneous flips in 5/6 pairs) and is excluded, following the stability criterion of [4]. Qwen, Gemma and Mistral showed 0 spontaneous flips in 1,200 runs each (95% upper bound on the per-run rate 0.25%).

**Target dependence (replication).** Consistent with [1, 3], $p_c$ depends on the pushed convention (Table 2); pushing Q from M is the hardest target in all three models and in DeepSeek live.

**Aggregation.** Averaging the *transition policy* across pairs is not the same as averaging *population outcomes* across pairs. The pooled emulator happens to land near the mixture's $p_c$ here (e.g. Qwen 0.272 vs 0.270), but its transition is far sharper: maximum flip-probability error 0.20–0.37 (Table 2). A mixture of sharp, target-specific transitions is broad, which matches the Test 1 failure pattern.

**Estimation.** We re-ran sampled extraction *in silico* with a **fixed, balanced target mixture** (K a multiple of 6: K/6 samples per pair and state), 30 replicates per K. The unsmoothed estimator's critical mass was nearly unbiased even at K = 6 (bias −0.005, +0.006 and −0.001 for Qwen, Mistral and Gemma; RMSE 0.013–0.023), whereas Jeffreys (bias −0.08 to −0.11) and Laplace (−0.15 to −0.20) estimates were strongly biased low at small K, with bias shrinking as K grows (Jeffreys −0.005 to +0.000 at K = 120). Interval calibration differed sharply (15 replicates each; nominal 90%): a parametric bootstrap from the unsmoothed estimate covered the true $p_c$ in 67–80% of replicates at K = 12 and 80–100% at K = 48, whereas Jeffreys posterior draws covered it in 0% at K = 12 and 33–53% at K = 48. Our preregistered predictive intervals were built from such draws, so they were shifted toward a lower critical mass. Unbiased estimates of each probability need not yield an unbiased estimate of the nonlinear threshold, and a zero count makes the unsmoothed estimate certain that a transition never happens. Neither smoothing nor Bayesian uncertainty propagation is ruled out; the specific posterior-mean estimators we tested are biased at small K.

Table 2: **Table 2.** Exact-policy ground truth (N = 24, 150-round horizon). Range of $p_c$ across six target conventions; maximum $|\Delta\varphi|$ of the pooled emulator relative to the counterbalanced mixture; bias of the critical-mass estimate at K = 6 samples per state (balanced design).

| Model | $p_c$ range across targets | Hardest target | Pooled max $\lvert\Delta\varphi\rvert$ | Bias at K = 6: MLE / Jeffreys / Laplace |
|---|---|---|---|---|
| Qwen3-8B | 0.23–0.38 | M → Q | 0.20 | −0.005 / −0.084 / −0.151 |

| Model | $p_c$ range across targets | Hardest target | Pooled max $\lvert\Delta\varphi\rvert$ | Bias at K = 6: MLE / Jeffreys / Laplace |
|---|---|---|---|---|
| Gemma-3-12B | 0.25–0.37 | M → Q | 0.20 | −0.001 / −0.112 / −0.198 |
| Mistral-7B | 0.07–0.41 | M → Q | 0.37 | +0.006 / −0.088 / −0.163 |

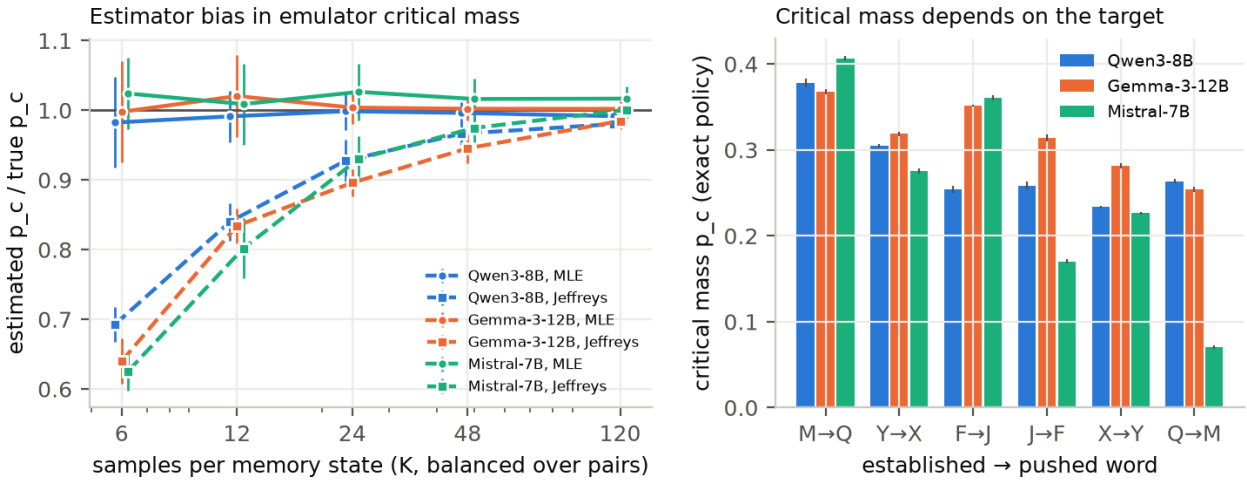



Figure 2: **Figure 2.** Exact-policy ground truth. *Left:* estimated / true critical mass vs samples per state in a balanced design (K/6 samples per word pair), mean ± SD over 30 replicates, for unsmoothed (MLE) and Jeffreys estimators. *Right:* critical mass by target convention (error bars: ±2 Monte Carlo SD over 8 seed sets; all ≤ 0.005).

## 4.4 Preregistered test 2: a target-conditional forecast

Using target-specific DeepSeek policies, we ran 35 new populations (Table 3, Fig. 3).

Table 3: **Table 3.** Preregistered test 2 (DeepSeek-v4-flash, 5 populations per cell). Values are listed per level of c.

| Direction | c | Live k/5 | Emulator φ | 95% PI (k) |
|---|---|---|---|---|
| push Q | 3 / 4 / 5 / 6 | 0 / 2 / 5 / 5 | 0.00 / 0.18 / 0.96 / 1.00 | [0,1] / [0,5] / [4,5] / [5,5] |
| push M | 2 / 3 / 4 | 1 / 5 / 5 | 0.01 / 0.59 / 1.00 | [0,4] / [1,5] / [5,5] |

**The directional forecast H5 is confirmed:** at c = 3–4, push M flipped 10/10 populations vs 2/10 for push Q (one-sided Fisher $p = 3.6\times10^{-4}$). It is robust to the event definition (100-round horizon: 8/10 vs 1/10, p = 0.003; sustained adoption: 10/10 vs 1/10, $p = 6\times10^{-5}$). **Quantitative calibration is only weakly tested:** H1′ passes (7/7 cells inside PIs), but intervals such as [0, 5] cannot be falsified with five populations. As proper scores, the target-conditional forecast reached Brier 0.094 [0.031, 0.173] (log-loss 0.33), better than a base-rate forecast (0.225; ΔBrier 95% CI [−0.224, −0.029]), whereas the pooled policy again matched the base rate (0.224). Time-to-tip equivalence was established in 1 of 4 eligible cells. With 10 vs 10 populations, H5 could detect only large effects (≈ 0.8 vs 0.2).

## 4.5 Monitoring: no consistent incremental value from the tested indicators

Pipeline validation and indicator calibration are in Appendix D.

**Unmatched design (as preregistered).** Level and trend alone separate TOWARD from null segments almost perfectly (CV AUROC 0.97–1.00), so H4 passes trivially and is uninformative.

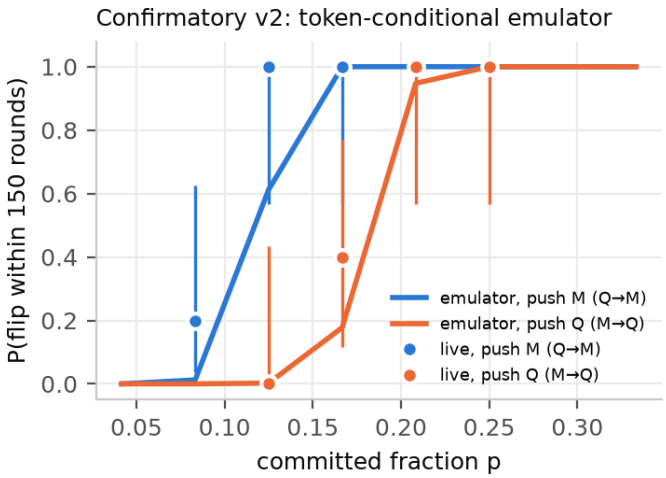



Figure 3: **Figure 3.** Preregistered test 2: live flip fraction (Wilson 95% CI) vs the target-conditional emulator, for both directions.

**Level-matched design (Fig. 4).** After matching, level and trend are uninformative (AUROC ≈ 0.50) while the oracle reaches 0.87–1.00 at leads of 25–50 rounds: the hidden driver carries information that is not in level or trend. Calibrated indicators alone reach 0.43–0.58 with no consistent sign across leads or policies. Their incremental value over level + trend ranged from −0.17 to +0.05 AUROC; with the cluster bootstrap, CIs are about ±0.05–0.06 and no calibrated indicator significantly helps, and with the refit bootstrap (training variability included) every interval contains zero but spans roughly ±0.08–0.12. H4 is therefore *not falsified* for the calibrated indicators, but equivalence within 0.02 is not established and gains of up to ≈0.1 AUROC cannot be excluded. The preregistered *raw* variance indicator formally fails H4 for Gemma at lead 50 (+0.099 [+0.045, +0.149]); its calibrated counterpart does not (+0.039 [−0.013, +0.089]), consistent with an indicator artefact. Matching reuses null runs (e.g. 239 matched pairs drew on 96 distinct null runs for Gemma at lead 25), which the cluster bootstrap accounts for. We conclude that **the tested indicators showed no consistent incremental discrimination**; this is not an information-theoretic statement about the defender's observations, and we did not evaluate prospective alarms (run-level false alarms and lead time). The monitoring emulations used pooled policies, which §4.3 shows can distort dynamics near threshold.

# 5. Discussion and Limitations

**For forecasting swarm vulnerability.** (1) For this sampled API model, live decisions track the extracted policy closely enough that the critical-mass forecast is essentially unchanged; this is a check of one model and one prompt, not a general guarantee. (2) The decisive errors were in *how* the policy was used: averaging transition policies across contexts produced a confidently wrong, too-sharp transition, and posterior-mean smoothing at small sample sizes made populations look easier to take over. Forecasts should be target-conditional, the estimator's bias and coverage should be checked against a known policy, and uncertainty should be propagated in a way that can represent unseen transitions. (3) Critical mass is a property of the (population, target) pair [1, 3]; the forecast that matters is for the target an adversary would push.

**For monitoring.** The tested indicators did not add discrimination beyond population state and trend; the oracle result points to the driver itself (e.g., integrity checks on agent memories) as the more promising signal.

### Limitations

*Scope.* One live API model; N = 24 live (N = 48 in monitoring emulations); 4–8 populations per level (100 requests/min quota); one abstract coordination game, memory length, temperature and label set; reasoning disabled. No claim about the Hugging Face incident beyond motivation. *Measurement.* DeepSeek policies are sampled; local policies are exact only for the forced-prefix two-answer policy of specific 4-bit builds. *Inference.* Preregistered PIs used biased Jeffreys draws; H1′ and H4 are weak; Test 1 pair allocation was unbalanced across levels; per-pair post-hoc analyses used extraction samples

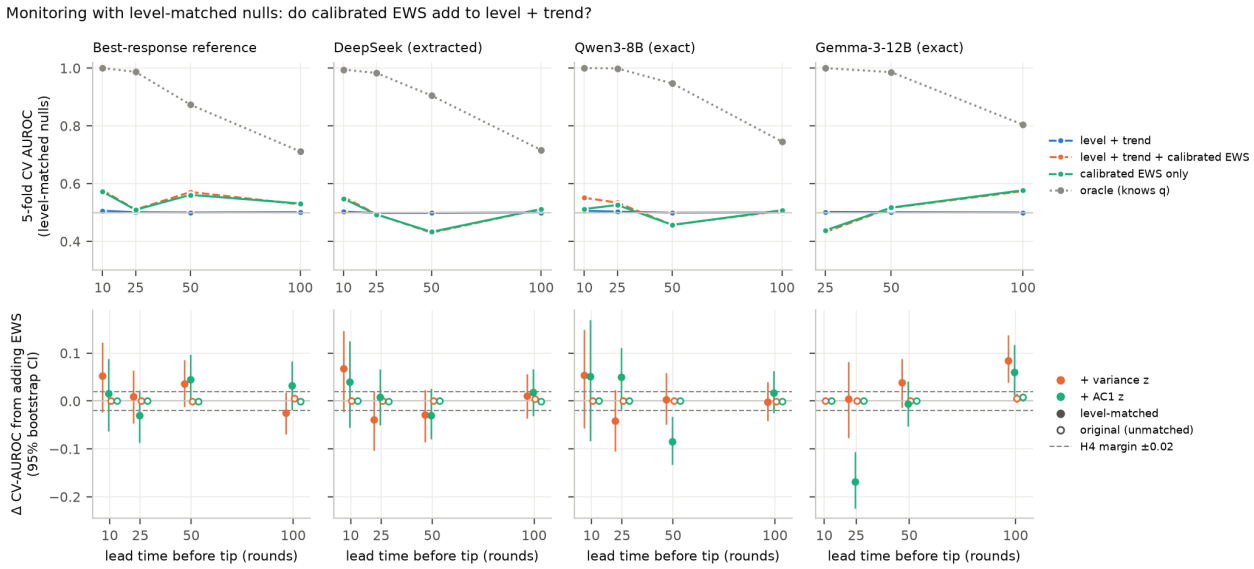



Figure 4: **Figure 4.** Level-matched monitoring (emulation, N = 48; four policies). *Top:* CV AUROC of level + trend, level + trend + calibrated indicators, indicators only, and an oracle that knows the poisoning rate. *Bottom:* ΔAUROC from adding each indicator (95% cluster-bootstrap CI over connected components of matched pairs); dashed lines mark the H4 margin ±0.02.

partly collected after Test 1 (independent of its outcomes). The policy-following check aggregates over option order in its primary analysis and cannot establish independence between calls. *Monitoring* is retrospective, emulated, with pooled policies and one driver.

## 6. Conclusion

Prospective tests separate what an emulator gets right from what it gets wrong. For one API model, live agents tracked their extracted policy closely enough that the critical-mass forecast was unchanged, yet a forecast built from a pooled policy failed and had no skill, while a target-conditional forecast confirmed a large asymmetry. Exact policies trace the failure to aggregation across contexts and to biased small-sample estimators. Early-warning indicators of critical slowing down did not add discrimination in our emulations. Forecasts of swarm vulnerability should be target-specific, estimator-checked, and tested prospectively. Next steps are a calibration-sized forecasting benchmark on fresh targets and models, active allocation of API queries to the states that most influence the forecast, single-agent probes that predict population critical mass, and prospective alarm evaluation.

## Code and Data

- **Code repository:** https://github.com/sharath-sms/Swarm-Alignment-Hackathon-Apart-Research (environment `swarm_ews/envs/naming_game.py`, emulator `swarm_ews/emulator.py`, analyses in `scripts/`, tests in `tests/`).
- **Data:** a sanitized decision-level dataset of all live decisions and extraction samples (`results/decisions/deepseek_decisions.` pair, option order, memory state, choice; no prompts or responses) lets §4.1 be rerun without the API cache; extracted and exact policies, predictions, run records and analysis outputs are in `results/`. Ashery et al. [1] data used under MIT licence.
- **Preregistrations:** documents and SHA-256 hashes are available from the author.

## Author Contributions

S.M.S. conceived the project, designed and ran all experiments, performed the analysis and wrote the report.

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# Appendix

## A. Exploratory analyses: withdrawn and downgraded claims

In the Ashery et al. Claude archive [1], a censored survival fit strongly prefers a threshold law over no-threshold dilution ($\Delta$AIC = 38; $p_c$ = 0.156 [0.144, 0.162]), and time-to-takeover rises steeply near the threshold (restricted mean to 800 interactions: 202 at c = 5, 464 at c = 4, 753 at c = 3). We initially reported an exponent $\gamma = 0.53$ as evidence for saddle-node scaling ($\gamma = ½$). **We withdraw this claim:** parameter recovery shows that the estimator returns 0.58–0.70 on emulators with known saddle-nodes, exact Qwen/Gemma policies give $\gamma \approx 0.86/1.05$, and large-N simulations show no single exponent (Fig. 5). In the same archive, Llama-3-70B (N = 48) holds its convention without adversaries but tips with a single committed agent (p = 0.02) in 16/21 runs; its threshold lies below the tested range. We also withdraw an earlier "less excess variance before tips" result, which came from the uncalibrated variance indicator.

## B. Dual-use and ethical considerations

**Dual use.** Target choice changes the number of compromised agents required, and memory poisoning tips populations; an attacker could use this to choose easier targets. We judge the marginal uplift low: the setting is an abstract coordination game, convention dependence of critical mass is already published [1, 3], and the practical findings are defensive (how to forecast vulnerability and check forecasts). We release a research simulator, not attack tooling. **Data and ethics.** No human subjects were involved. The Flint et al. group-size repository (no licence file) was not used.

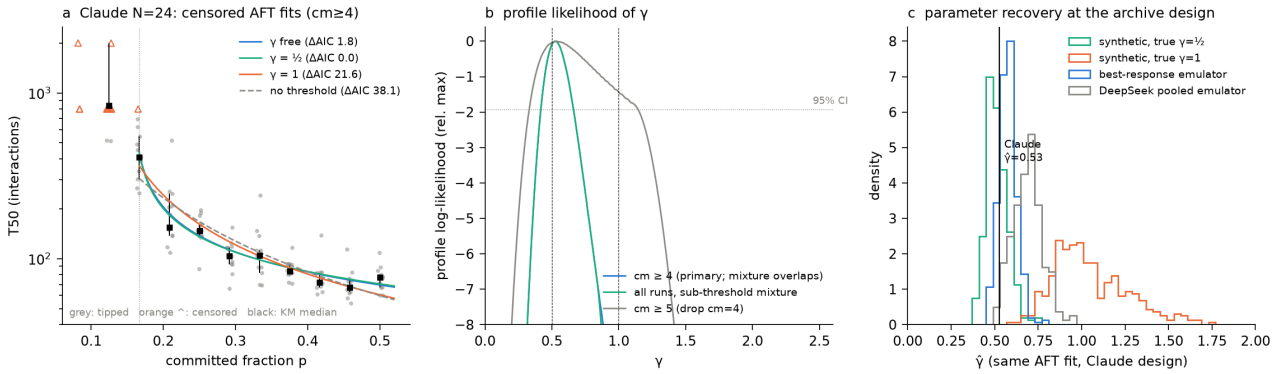



Figure 5: **Figure 5.** Claude archive (exploratory). *a:* censored accelerated-failure-time fits; *b:* profile likelihood of γ; *c:* parameter recovery at the archive design.

## C. Research integrity

All failed preregistered predictions (H1 in Test 1; P5; raw-variance H4 for Gemma) and withdrawn claims (γ = ½; reduced pre-tip variance; an earlier uncalibrated "statistically indistinguishable" agreement claim) are reported. An independent internal critique, an external review and our responses are available from the author. All code, data collection and analysis were done during the sprint (6–8 November 2026); the only pre-existing assets are the public Ashery et al. dataset and code (MIT), whose prompt we adapted.

## D. Monitoring pipeline validation

**Pipeline validation.** On synthetic systems the pipeline detects a fold against a mean-drift null (AUROC 0.95) and does not fire on mean drift alone (0.54). A preregistered check (P5: a level-only baseline cannot separate fold from mean drift) **failed** (0.62). The raw binomial-normalised variance is biased on pure noise when the mean moves (Kendall τ up to +0.4–0.7), so we recalibrated it with surrogates; residual bias remains when the mean accelerates faster than the detrending window, i.e. just before a tip.

# LLM Usage Statement

Claude was used for literature search, coding, analysis, drafting and editing of this report. All results and claims were checked by the author against the experiment outputs, and the author takes full responsibility for the content.